

Dear Hiring Manager,

I am excited to apply for the electrical engineering intern position at Collins Aerospace. As a sophomore majoring in Computer Engineering at the University of Maryland, College Park, I am eager to leverage my passion for problem-solving and my growing expertise in hardware and software to contribute to Collins Aerospace's mission of pushing the boundaries of known science to solve the world's most complex problems.

During my internship at the Adelphi Lab Center in the U.S. Army Combat Capabilities Development Command, I was tasked with converting a Field Fox handheld microwave analyzer into a radar system in order to provide a cost effective implementation. To do so, I made extensive use of MATLAB and Standard Commands for Programmable Instruments to interface with the Field Fox. I also designed and implemented a test procedure that began with indoor experiments using calibration targets and progressed to outdoor trials for improved background subtraction.

Previously, at the Johns Hopkins Applied Physics Laboratory (JHUAPL) I completed two internships that deepened my experience in computer vision and simulation. For NASA's Dragonfly mission, I compared MATLAB's Camera Calibrator and OpenCV's Camera Calibrator software. I then tested both software in a MATLAB simulation that used previous Dragonfly prototype test flight data. Due to my contributions, the MATLAB Camera Calibrator is now used to calibrate Dragonfly's prototype camera before test flights. In my following project, I worked to train a computer vision model to detect lunar craters. I developed a dataset of synthetic lunar images. I then inputted the training data into the YOLOv8 computer vision model and tested the trained model on real-life images of lunar craters using Python, significantly improving the YOLOv8 model's crater detection accuracy.

Together, these experiences strengthened my problem solving skills and deepened my interest in engineering. With a blend of hardware integration and software development skills, I am well prepared to contribute meaningfully as an electrical engineering intern at Collins Aerospace.

Thank you for considering my application. I look forward to the possibility of discussing how my background, skills, and enthusiasm align with the exciting work at Collins Aerospace.

Sincerely,  
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